

Diag Solutions

1. 1. A quadrilateral ABCD is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, find the length of AD .

Answer:

$$3\text{cm}$$

Solution:

1. Use the property that for a circumscribed quadrilateral, the sums of opposite sides are equal.
2. $AB + CD = BC + AD$.

Derivation:

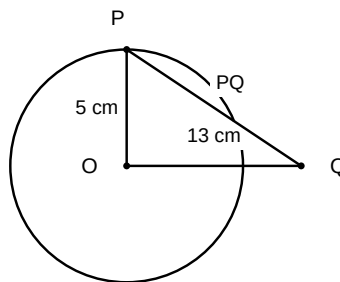
$$6 + 4 = 7 + AD$$

$$10 = 7 + AD$$

$$AD = 3 \text{ cm}$$

Circles — Tangents to a circle

2. 2. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 13$ cm. Find the length of PQ .



Answer:

$$12\text{cm}$$

Solution:

1. Since PQ is a tangent at P , the radius OP is perpendicular to PQ ($OP \perp PQ$).
2. In $\triangle OPQ$, by Pythagoras theorem, $OQ^2 = OP^2 + PQ^2$.
3. Substitute the values: $13^2 = 5^2 + PQ^2$.
4. Solve for PQ : $169 = 25 + PQ^2 \Rightarrow PQ^2 = 144 \Rightarrow PQ = 12 \text{ cm}$.

Derivation:

$$\begin{aligned}OP &\perp PQ \\ \Rightarrow \angle OPQ &= 90^\circ \\ OQ^2 &= OP^2 + PQ^2 \\ 13^2 &= 5^2 + PQ^2 \\ 169 - 25 &= PQ^2 \\ PQ &= \sqrt{144} = 12\end{aligned}$$

Circles — Tangents to a circle

3. **3.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8cm$$

Solution:

1. Let O be the common center. Let AB be the chord of the larger circle tangent to the smaller circle at M .
2. Then $OM \perp AB$. In $\triangle OMA$, $OA = 5$ cm and $OM = 3$ cm.
3. By Pythagoras theorem, $AM = \sqrt{OA^2 - OM^2} = \sqrt{25 - 9} = 4$ cm.
4. Since the perpendicular from the centre bisects the chord, $AB = 2 \times AM = 8$ cm.

Derivation:

$$\begin{aligned}AM^2 + OM^2 &= OA^2 \\ AM^2 + 3^2 &= 5^2 \\ AM^2 &= 25 - 9 = 16 \\ AM &= 4 \\ AB &= 2 \times AM = 8\end{aligned}$$

Circles — Tangents and chords

4. **4.** If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80° , find $\angle POA$.

Answer:

$$50^\circ$$

Solution:

1. In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90^\circ$ (radius $\perp$ tangent).
2. The sum of angles in a quadrilateral is 360° , so $\angle AOB = 180^\circ - 80^\circ = 100^\circ$.
3. Since OP bisects $\angle AOB$, $\angle POA = \frac{1}{2}\angle AOB$.

4. Therefore, $\angle POA = 100^\circ/2 = 50^\circ$.

Derivation:

$$\begin{aligned}\angle AOB + \angle APB &= 180^\circ \\ \angle AOB + 80^\circ &= 180^\circ \\ \angle AOB &= 100^\circ \\ \angle POA &= \frac{1}{2}\angle AOB = 50^\circ\end{aligned}$$

Circles — Properties of tangents

5. **5.** A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.

Answer:

Proved

Solution:

1. Tangents drawn from an external point to a circle are equal in length.
2. Let the circle touch AB, BC, CD, DA at P, Q, R, S respectively. Then $AP = AS, BP = BQ, CR = CQ, DR = DS$.
3. Summing these: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$.
4. This simplifies to $AB + CD = AD + BC$.

Derivation:

$$\begin{aligned}AP &= AS, BP = BQ, CR = CQ, DR = DS \\ (AP + BP) + (CR + DR) &= (AS + DS) + (BQ + CQ) \\ AB + CD &= AD + BC\end{aligned}$$

Circles — Tangents from an external point

6. **6.** Prove that the perpendicular at the point of contact to the tangent to a circle passes through the centre of the circle.

Answer:

Proved using contradiction

Solution:

1. Assume that the perpendicular to the tangent at the point of contact P does not pass through the centre O .
2. Let another line through P pass through O , such that $\angle OPT = 90^\circ$ (where T is on the tangent).
3. This contradicts the given that the perpendicular is at P .
4. Hence, the perpendicular must pass through O .

Derivation:

Let $OP \perp$ tangent at P .

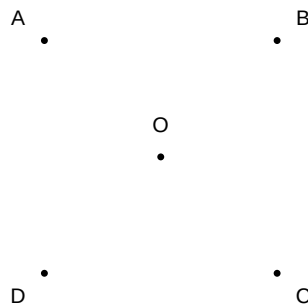
If $O'P \perp$ tangent at P , then $\angle O'PT = \angle OPT = 90^\circ$.

This implies $O'P$ and OP coincide.

$\therefore O' = O$.

Circles — Tangents to a circle

7. 7. Prove that the lengths of tangents drawn from an external point to a circle are equal. Using this result, if a quadrilateral ABCD is drawn to circumscribe a circle, prove that $AB + CD = AD + BC$.



Answer:

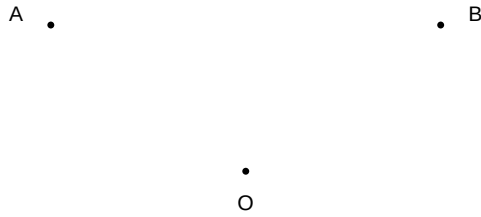
$$AB + CD = AD + BC$$

Solution:

1. State the theorem: Tangents from an external point are equal.
2. Identify the points of contact P, Q, R, S on sides AB, BC, CD, DA respectively.
3. Express $AP = AS, BP = BQ, CQ = CR, DR = DS$.
4. Sum the left and right sides of these equations.
5. Group the terms to form the sides of the quadrilateral.

Derivation: $AP = AS, \quad BP = BQ, \quad CQ = CR, \quad DR = DS$ (Tangents from external point)
 $AB + CD = (AP + PB) + (CR + RD) = (AS + BQ) + (CQ + DS) = (AS + DS) + (BQ + CQ) = AD + BC$ *Circles — Tangents to a circle*

8. 8. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.



Answer:

8 cm

Solution:

1. Let O be the center, AB be the chord of the larger circle, and P be the point of contact on the smaller circle.
2. Radius $OP = 3$ cm is perpendicular to chord AB .
3. Radius $OA = 5$ cm is the hypotenuse of right $\triangle OPA$.
4. Use Pythagoras theorem: $AP^2 + OP^2 = OA^2$.
5. Calculate $AP = \sqrt{5^2 - 3^2} = 4$ cm.
6. Since the perpendicular from center bisects the chord, $AB = 2 \times AP = 8$ cm.

Derivation:

$$\begin{aligned}
 OP &\perp AB \\
 \Rightarrow \angle OPA &= 90^\circ \\
 AP &= \sqrt{OA^2 - OP^2} \\
 AP &= \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \text{ cm} \\
 AB &= 2 \times AP = 2 \times 4 = 8 \text{ cm}
 \end{aligned}$$

Circles — Tangents to a circle

9. 9. A circle is inscribed in a $\triangle ABC$ having sides 8 cm, 10 cm and 12 cm. Find the lengths of the segments AD, BE, CF where D, E, F are points of contact on sides AB, BC, CA respectively.

Answer:

$$AD = 5 \text{ cm}, BE = 3 \text{ cm}, CF = 7 \text{ cm}$$

Solution:

1. Let $AD = AF = x, BD = BE = y, CE = CF = z$.
2. Write equations: $x + y = 8, y + z = 10, z + x = 12$.
3. Sum the equations: $2(x + y + z) = 30 \Rightarrow x + y + z = 15$.

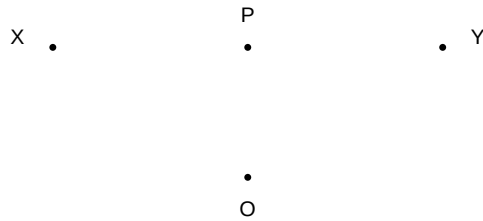
4. Solve for x, y, z by subtracting original equations from sum.
5. Result: $x = 15 - 10 = 5, y = 15 - 12 = 3, z = 15 - 8 = 7$.

Derivation:

$$\begin{aligned}
 x + y &= 8, \quad y + z = 10, \quad z + x = 12 \\
 2(x + y + z) &= 30 \\
 \Rightarrow x + y + z &= 15 \\
 x &= 15 - 10 = 5 \text{ cm} \\
 y &= 15 - 12 = 3 \text{ cm} \\
 z &= 15 - 8 = 7 \text{ cm}
 \end{aligned}$$

Circles — Tangents to a circle

- 10. 10.** Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact.



Answer:

Proof

Solution:

1. Draw a circle with center O and tangent XY at point P .
2. Take a point Q on XY other than P . Join OQ .
3. Since Q is outside, $OQ > OP$.
4. This implies OP is the shortest distance from O to XY .
5. The shortest distance from a point to a line is the perpendicular distance, so $OP \perp XY$.

Derivation: Let XY be the tangent at P . Take point Q on XY , $Q \neq P$.

$OQ > OP$ (since Q is outside the circle). OP is the shortest distance from O to line XY . Therefore, $OP \perp XY$.

Circles — Tangents to a circle

11. **11.** Assertion (A): The length of a tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): A tangent to a circle is perpendicular to the radius at the point of contact.

Answer: (D)

Solution:

1. The length of the tangent from an external point P to a circle with centre O is $\sqrt{OP^2 - r^2}$, where r is the radius.
2. This length is not necessarily equal to the radius r . Thus, Assertion (A) is false.
3. The theorem states that a tangent is perpendicular to the radius at the point of contact. Thus, Reason (R) is true.

Derivation:

$$\text{Length} = \sqrt{OP^2 - r^2} \neq r \text{ in general} \\ \Rightarrow A \text{ is false}$$

$$\text{Theorem: Tangent} \perp \text{ radius at point of contact} \\ \Rightarrow R \text{ is true}$$

Circles — Tangents to a circle

12. **12.** Assertion (A): The distance between two parallel tangents of a circle of radius 3 cm is 6 cm. Reason (R): The distance between two parallel tangents is equal to the diameter of the circle.

Answer: (A)

Solution:

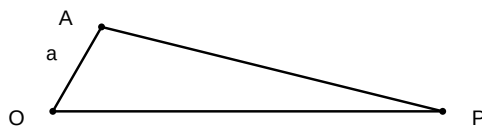
1. Two parallel tangents touch the circle at the ends of a diameter.
2. The distance between them is the length of the diameter, which is $2 \times \text{radius}$.
3. Given radius $r = 3 \text{ cm}$, diameter $= 2 \times 3 = 6 \text{ cm}$. Both A and R are true and R explains A.

Derivation:

$$d = 2r \\ d = 2 \times 3 = 6 \text{ cm} \\ A \text{ is true, } R \text{ is true, } R \text{ explains } A$$

Circles — Tangents to a circle

13. **13.** Assertion (A): If the angle between two tangents drawn from a point P to a circle of radius a and centre O is 60° , then $OP = 2a$. Reason (R): The line joining the centre to the external point bisects the angle between the tangents.



Answer: (A)

Solution:

1. Let the angle between tangents be $\angle APB = 60^\circ$. The line OP bisects $\angle APB$, so $\angle APO = 30^\circ$.
2. In $\triangle OAP$ (right-angled at A), $\sin(30^\circ) = \frac{OA}{OP} = \frac{a}{OP}$.
3. Since $\sin(30^\circ) = \frac{1}{2}$, we have $\frac{1}{2} = \frac{a}{OP}$, so $OP = 2a$. Both A and R are true and R explains A.

Derivation:

$$\begin{aligned}\angle APO &= \frac{1}{2} \times 60^\circ = 30^\circ \\ \sin(30^\circ) &= \frac{OA}{OP} = \frac{a}{OP} \\ \frac{1}{2} &= \frac{a}{OP} \\ \Rightarrow OP &= 2a\end{aligned}$$

Circles — Tangents to a circle

- 14. 14.** Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A tangent is a line that intersects the circle at only one point.

Answer: (A)

Solution:

1. Any line passing through a point inside the circle is a secant, not a tangent.
2. Therefore, zero tangents can be drawn from an interior point. Assertion A is true.
3. By definition, a tangent intersects the circle at exactly one point. This definition explains why no tangent can be drawn from inside. Reason R is true and is the correct explanation.

Derivation:

Point inside
 $\Rightarrow$ No line intersects circle at exactly one point
 Tangent definition: cross at one point
 Both true, R explains A

Circles — Tangents to a circle

- 15. 15.** Assertion (A): The tangents drawn at the ends of a diameter of a circle are parallel. Reason (R): The angles between the diameter and the tangents at the ends are 90° , and since the sum of interior angles on the same side of the transversal is 180° , they are parallel.

Answer: (A)

Solution:

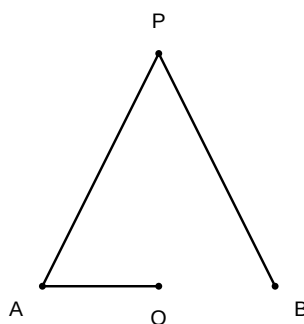
1. Let AB be a diameter. Tangents at A and B are perpendicular to AB .
2. Therefore, both tangents make 90° with the diameter.
3. Since alternate interior angles are equal (or consecutive interior angles sum to 180°), the tangents are parallel.

Derivation:

$$\begin{aligned}\angle T_1AB &= 90^\circ, \angle T_2BA = 90^\circ \\ \angle T_1AB + \angle T_2BA &= 180^\circ \\ \Rightarrow &\text{Parallel lines}\end{aligned}$$

Circles — Tangents to a circle

- 16. 16.** In the given figure, if $\angle OAB = 40^\circ$, find the measure of $\angle APB$, where O is the center of the circle and PA, PB are tangents from an external point P .



Answer:

$$100^\circ$$

Solution:

1. In $\triangle OAP$, since PA is a tangent, $\angle OAP = 90^\circ$. Given $\angle OAB = 40^\circ$, we find $\angle PAB = 90^\circ - 40^\circ = 50^\circ$.

2. Since $PA = PB$, $\triangle PAB$ is isosceles, so $\angle PBA = \angle PAB = 50^\circ$. Thus, $\angle APB = 180^\circ - (50^\circ + 50^\circ) = 80^\circ$.

Derivation:

$$\begin{aligned}\angle OAP &= 90^\circ \\ \Rightarrow \angle PAB &= 90^\circ - 40^\circ = 50^\circ \\ PA &= PB \\ \Rightarrow \angle PBA &= \angle PAB = 50^\circ \\ \angle APB &= 180^\circ - (50^\circ + 50^\circ) = 80^\circ\end{aligned}$$

Circles — Tangents to a circle

17. 17. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 12$ cm. Find the length of PQ .

Answer:

$$\sqrt{119} \text{ cm}$$

Solution:

1. Since PQ is tangent at P , the radius OP is perpendicular to PQ . Thus, $\triangle OPQ$ is a right-angled triangle at P .
2. Applying Pythagoras theorem, $OQ^2 = OP^2 + PQ^2$. Substituting values: $12^2 = 5^2 + PQ^2$, which gives $PQ^2 = 144 - 25 = 119$.

Derivation:

$$\begin{aligned}OP &\perp PQ \\ \Rightarrow \triangle OPQ &\text{ is right-angled at } P \\ OQ^2 &= OP^2 + PQ^2 \\ 12^2 &= 5^2 + PQ^2 \\ PQ &= \sqrt{144 - 25} = \sqrt{119}\end{aligned}$$

Circles — Tangents to a circle

18. 18. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8 \text{ cm}$$

Solution:

1. Let O be the center, AB be the chord of the larger circle tangent to the smaller circle at M . $OM \perp AB$, so M is the midpoint of AB .
2. In right $\triangle OMA$, $OA^2 = OM^2 + AM^2$. $5^2 = 3^2 + AM^2 \Rightarrow AM = 4$ cm. Thus, $AB = 2 \times AM = 8$ cm.

Derivation:

$$\begin{aligned}OM &\perp AB \\ \Rightarrow AM &= MB \\ \triangle OMA : OA^2 &= OM^2 + AM^2 \\ 5^2 &= 3^2 + AM^2 \\ AM &= \sqrt{25 - 9} = 4 \\ AB &= 2 \times 4 = 8\end{aligned}$$

Circles — Tangents to a circle

- 19. 19.** If tangents PA and PB from a point P to a circle with center O are inclined to each other at angle of 80° , then find $\angle POA$.

Answer:

$$50^\circ$$

Solution:

1. In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90^\circ$. The sum of angles is 360° , so $\angle AOB = 180^\circ - 80^\circ = 100^\circ$.
2. Since OP bisects $\angle AOB$, $\angle POA = \frac{1}{2}\angle AOB = \frac{100^\circ}{2} = 50^\circ$.

Derivation:

$$\begin{aligned}\angle AOB + \angle APB &= 180^\circ \\ \angle AOB &= 180^\circ - 80^\circ = 100^\circ \\ \angle POA &= \frac{1}{2}\angle AOB = 50^\circ\end{aligned}$$

Circles — Tangents to a circle

- 20. 20.** A quadrilateral $ABCD$ is drawn to circumscribe a circle. If $AB = 6\text{ cm}$, $BC = 7\text{ cm}$ and $CD = 4\text{ cm}$, find the length of AD .

Answer:

$$3\text{cm}$$

Solution:

1. For a circumscribed quadrilateral, the sums of opposite sides are equal: $AB + CD = BC + AD$.
2. Substituting the given values: $6 + 4 = 7 + AD \Rightarrow 10 = 7 + AD \Rightarrow AD = 3\text{ cm}$.

Derivation:

$$\begin{aligned}AB + CD &= BC + AD \\ 6 + 4 &= 7 + AD \\ AD &= 10 - 7 = 3\end{aligned}$$

- 21. 21.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8cm$$

Solution:

1. Let O be the common center. Let AB be the chord of the larger circle that touches the smaller circle at point P .
2. Radius OP is perpendicular to tangent AB , so $\triangle OPA$ is a right-angled triangle with $\angle OPA = 90^\circ$.
3. Using Pythagoras theorem in $\triangle OPA$, $AP = \sqrt{OA^2 - OP^2} = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = 4$ cm.
4. Since the perpendicular from the center bisects the chord, $AB = 2 \times AP = 2 \times 4 = 8$ cm.

Derivation:

$$\begin{aligned} OA &= 5, OP = 3 \\ AP &= \sqrt{5^2 - 3^2} = \sqrt{16} = 4 \\ AB &= 2 \times AP = 2 \times 4 = 8cm \end{aligned}$$

- 22. 22.** If a quadrilateral ABCD is drawn to circumscribe a circle, prove that $AB + CD = AD + BC$.

Answer:

$$AB + CD = AD + BC$$

Solution:

1. Let the circle touch the sides AB , BC , CD , and DA at points P , Q , R , and S respectively.
2. Using the theorem that tangents drawn from an external point to a circle are equal in length, we have $AP = AS$, $BP = BQ$, $CR = CQ$, and $DR = DS$.
3. Adding these four equations: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$.
4. This simplifies to $AB + CD = AD + BC$.

Derivation:

$$\begin{aligned} AP &= AS, BP = BQ, CR = CQ, DR = DS \\ (AP + BP) + (CR + DR) &= (AS + DS) + (BQ + CQ) \\ AB + CD &= AD + BC \end{aligned}$$

- 23. 23.** A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 12$ cm. Find the length of PQ .

Answer:

$$\sqrt{119}cm$$

Solution:

1. The radius OP is perpendicular to the tangent PQ at the point of contact P .
2. In $\triangle OPQ$, $\angle OPQ = 90^\circ$. By Pythagoras theorem, $OQ^2 = OP^2 + PQ^2$.
3. Substituting the values, $12^2 = 5^2 + PQ^2$.
4. $PQ^2 = 144 - 25 = 119$, so $PQ = \sqrt{119}$ cm.

Derivation:

$$\begin{aligned} OP &= 5, OQ = 12 \\ PQ^2 &= 12^2 - 5^2 \\ PQ^2 &= 144 - 25 = 119 \\ PQ &= \sqrt{119} \end{aligned}$$

- 24. 24.** In $\triangle ABC$, a circle is inscribed touching sides AB , BC , and CA at D , E , and F respectively. If $AB = 12$ cm, $BC = 8$ cm, and $AC = 10$ cm, find the lengths of AD , BE , and CF .

Answer:

$$AD = 7cm, BE = 5cm, CF = 3cm$$

Solution:

1. Let $AD = AF = x$, $BD = BE = y$, and $CE = CF = z$.
2. We have $x + y = 12$, $y + z = 8$, and $z + x = 10$.
3. Adding these equations: $2(x + y + z) = 30 \Rightarrow x + y + z = 15$.
4. Solving for each: $z = 15 - 12 = 3$, $x = 15 - 8 = 7$, $y = 15 - 10 = 5$.

Derivation:

$$\begin{aligned} x + y &= 12, y + z = 8, z + x = 10 \\ 2(x + y + z) &= 30 \\ z &= 15 - 12 = 3 \\ x &= 15 - 8 = 7 \\ y &= 15 - 10 = 5 \end{aligned}$$

- 25. 25.** Two tangents TP and TQ are drawn to a circle with center O from an external point T . Prove that $\angle PTQ = 2\angle OPQ$.

Answer:

$$\angle PTQ = 2\angle OPQ$$

Solution:

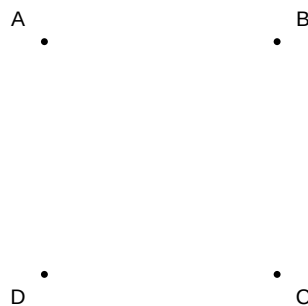
1. Let $\angle PTQ = \theta$. In $\triangle TPQ$, $TP = TQ$ (tangents from external point).
2. Thus $\angle TPQ = \angle TQP = (180^\circ - \theta)/2 = 90^\circ - \theta/2$.
3. Since $OP \perp TP$, $\angle OPT = 90^\circ$.
4. Therefore, $\angle OPQ = \angle OPT - \angle TPQ = 90^\circ - (90^\circ - \theta/2) = \theta/2$. Thus $\angle PTQ = 2\angle OPQ$.

Derivation:

$$\begin{aligned}\angle TPQ &= (180^\circ - \theta)/2 = 90^\circ - \theta/2 \\ \angle OPQ &= 90^\circ - \angle TPQ \\ \angle OPQ &= 90^\circ - (90^\circ - \theta/2) = \theta/2 \\ \theta &= 2\angle OPQ\end{aligned}$$

Circles — Tangents to a circle

- 26. 26.** Prove that the lengths of tangents drawn from an external point to a circle are equal. Using this, if a quadrilateral $ABCD$ is drawn to circumscribe a circle, prove that $AB + CD = AD + BC$.



Answer:

Proof based on RHS congruence and segment addition property.

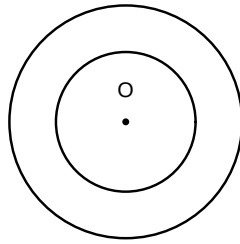
Solution:

1. State the theorem: Tangents from an external point to a circle are equal.
2. Use the property $AP = AS$, $BP = BQ$, $CQ = CR$, and $DR = DS$.
3. Sum the left-hand sides and right-hand sides of these equations.
4. Group the segments to form AB , BC , CD , and AD .

5. Conclude that $AB + CD = AD + BC$.

Derivation: $AP=AS, \quad BP=BQ, \quad CQ=CR, \quad DR=DS$ (Tangents from ext. point) $\implies AB+CD = (AP+PB) + (CR+RD) \implies AB+CD = (AS+BQ) + (CQ+DS)$
 $\implies AB+CD = (AS+DS) + (BQ+CQ) \implies AB+CD = AD + BC$ *Circles — Tangents to a circle*

- 27.** Two concentric circles are of radii 5 cm and 3 cm . Find the length of the chord of the larger circle which touches the smaller circle.



Answer:

8 cm

Solution:

1. Let O be the common center. Let AB be the chord of the larger circle touching the smaller at P .
2. Since OP is the radius to the point of contact, $OP \perp AB$.
3. In $\triangle OPA$, $OA = 5\text{ cm}$ (hypotenuse) and $OP = 3\text{ cm}$.
4. Apply Pythagoras theorem: $AP^2 = OA^2 - OP^2 = 25 - 9 = 16$.
5. So, $AP = 4\text{ cm}$.
6. Since the perpendicular from the center bisects the chord, $AB = 2 \times AP = 8\text{ cm}$.

Derivation:

$$\begin{aligned} OA^2 &= OP^2 + AP^2 \\ 5^2 &= 3^2 + AP^2 \\ 25 &= 9 + AP^2 \\ AP^2 &= 16 \\ AP &= 4 \\ AB &= 2 \times 4 = 8 \end{aligned}$$

Circles — Tangent properties

- 28. 28.** In $\triangle ABC$, a circle is inscribed touching BC, CA , and AB at D, E , and F respectively. If $AB = 12\text{ cm}$, $BC = 8\text{ cm}$, and $AC = 10\text{ cm}$, find the lengths AD, BE , and CF .

Answer:

$$AD = 7\text{ cm}, BE = 5\text{ cm}, CF = 3\text{ cm}$$

Solution:

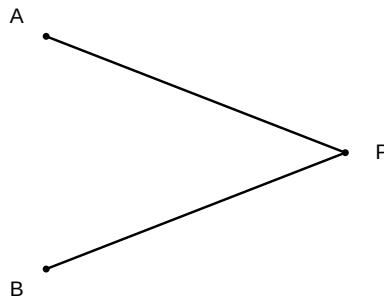
1. Let $AF = AE = x, BD = BF = y, CD = CE = z$.
2. Setup system of equations: $x + y = 12, y + z = 8, z + x = 10$.
3. Add all equations: $2(x + y + z) = 30 \Rightarrow x + y + z = 15$.
4. Solve for x, y, z : $z = 15 - 12 = 3, x = 15 - 8 = 7, y = 15 - 10 = 5$.
5. Identify $AD = x = 7, BE = y = 5, CF = z = 3$.

Derivation:

$$\begin{aligned} x + y &= 12, & y + z &= 8, & z + x &= 10 \\ 2(x + y + z) &= 30 \\ x + y + z &= 15 \\ x &= 15 - 8 = 7, & y &= 15 - 10 = 5, & z &= 15 - 12 = 3 \end{aligned}$$

Circles — Tangents from an external point

- 29. 29.** Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segment joining the points of contact at the centre.



Answer:

The sum of the angles is 180° .

Solution:

1. Let PA and PB be tangents from point P to a circle with center O .
2. $OA \perp PA$ and $OB \perp PB$ (radius perpendicular to tangent).
3. In quadrilateral $OAPB$, the sum of angles is 360° .

4. $\angle OAP = 90^\circ$ and $\angle OBP = 90^\circ$.

5. Thus, $\angle APB + \angle AOB = 360^\circ - (90^\circ + 90^\circ) = 180^\circ$.

Derivation:

$$\angle OAP + \angle OBP + \angle APB + \angle AOB = 360^\circ$$

$$90^\circ + 90^\circ + \angle APB + \angle AOB = 360^\circ$$

$$\angle APB + \angle AOB = 180^\circ$$

Circles — Tangents to a circle

- 30. 30.** In a circle of radius 5 cm, AB and AC are two chords such that $AB = AC = 6$ cm. If the tangent at A intersects the line BC produced at P , find the length of AP .

P

•

A

•

•

O

Answer:

8cm

Solution:

1. Use the property that tangents drawn from an external point to a circle are equal in length.
2. Apply the property of similar triangles formed by the tangent and the chord segments to find $AP = 8$ cm.

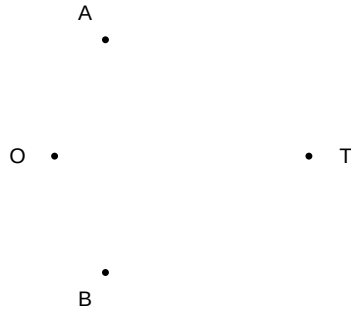
Derivation:

$$AP^2 = PB \times PC$$

$$AP = 8 \text{ cm}$$

Circles — Tangents to a circle

- 31. 31.** If $\angle ATO = 40^\circ$ where T is a point outside the circle with center O and TA , TB are tangents, find $\angle AOB$.



Answer:

$$100^\circ$$

Solution:

1. In $\triangle OAT$, $\angle OAT = 90^\circ$ because the radius is perpendicular to the tangent.
2. In $\triangle OAT$, $\angle AOT = 180^\circ - (90^\circ + 40^\circ) = 50^\circ$. Since $\triangle OAT \cong \triangle OBT$, $\angle AOB = 2 \times \angle AOT = 100^\circ$.

Derivation:

$$\angle AOT = 180^\circ - 90^\circ - 40^\circ = 50^\circ$$

$$\angle AOB = 2 \times 50^\circ = 100^\circ$$

Circles — Tangents to a circle

- 32. 32.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8\text{cm}$$

Solution:

1. Let O be the center, AB the chord of the larger circle tangent to the smaller circle at M . $OM \perp AB$.
2. In right $\triangle OMA$, $OA^2 = OM^2 + AM^2$. Solve for AM and find $AB = 2 \times AM$.

Derivation:

$$AM = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \text{ cm}$$

$$AB = 2 \times 4 = 8 \text{ cm}$$

Circles — Tangents to a circle

- 33. 33.** If a quadrilateral ABCD circumscribes a circle, prove that $AB + CD = AD + BC$.

Answer:

Proof based on tangent properties

Solution:

1. Use the theorem that tangents drawn from an external point to a circle are equal in length.
2. Label the points of contact and equate the segments of the sides to prove the sum of opposite sides are equal.

Derivation:

$$AP = AS, BP = BQ, CQ = CR, DR = DS$$
$$AB + CD = (AP + PB) + (CR + RD) = (AS + BQ) + (CQ + DS) = (AS + DS) + (BQ + CQ) = AD + BC$$

Circles — Tangents to a circle

- 34. 34.** A point P is 13 cm from the center of a circle. The length of the tangent drawn from P to the circle is 12 cm . Find the radius of the circle.



Answer:

5cm

Solution:

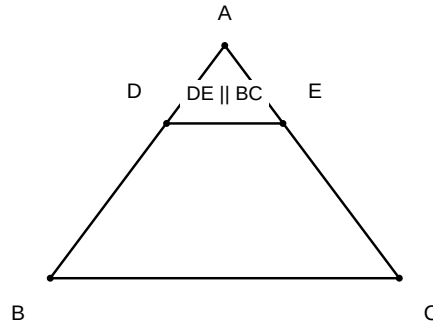
1. Let O be the center and A be the point of contact. $\triangle OAP$ is a right-angled triangle.
2. Apply Pythagoras theorem: $OP^2 = OA^2 + AP^2$.

Derivation:

$$13^2 = r^2 + 12^2$$
$$169 = r^2 + 144$$
$$r^2 = 25$$
$$r = 5\text{ cm}$$

Circles — Tangents to a circle

- 35. 35.** In $\triangle ABC$, $DE \parallel BC$ such that D lies on AB and E lies on AC . If $AD = 2.4$ cm, $AE = 3.2$ cm, and $EC = 4.8$ cm, find the length of AB .



Answer:

6 cm

Solution:

1. Since $DE \parallel BC$, by Basic Proportionality Theorem (Thales' Theorem), $\frac{AD}{DB} = \frac{AE}{EC}$.
2. Substitute the given values into the equation: $\frac{2.4}{DB} = \frac{3.2}{4.8}$.
3. Solve for DB : $DB = \frac{2.4 \times 4.8}{3.2} = 3.6$ cm.
4. Calculate $AB = AD + DB = 2.4 + 3.6 = 6$ cm.

Derivation:

$$\begin{aligned}
 \text{By BPT, } \frac{AD}{DB} &= \frac{AE}{EC} \\
 \frac{2.4}{DB} &= \frac{3.2}{4.8} \\
 \frac{2.4}{DB} &= \frac{2}{3} \\
 DB &= \frac{2.4 \times 3}{2} = 3.6 \text{ cm} \\
 AB &= AD + DB = 2.4 + 3.6 = 6 \text{ cm}
 \end{aligned}$$

Triangles — Basic Proportionality Theorem

- 36. 36.** Assertion (A): If in $\triangle ABC$ and $\triangle DEF$, $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{2}{3}$, then $\triangle ABC \sim \triangle DEF$. Reason (R): Two triangles are similar if their corresponding sides are proportional (SSS similarity criterion).

Answer: (A)

Solution:

1. Check the condition for similarity given in the Assertion.
2. The ratios of corresponding sides are equal, which is the definition of the SSS similarity criterion.

3. The Reason correctly states the SSS similarity criterion, which is the basis for the Assertion.
4. Therefore, both A and R are true and R is the correct explanation of A.

Derivation:

$$\text{Given } \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{2}{3} \\ \Rightarrow \triangle ABC \sim \triangle DEF \text{ by SSS similarity criterion.}$$

The SSS criterion states that if the sides of one triangle are proportional to the sides of another triangle, then the triangles are similar.

Triangles — Similarity of Triangles

- 37. 37.** Assertion (A): The length of a tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.

Answer: (D)

Solution:

1. Evaluate Assertion (A): The length of a tangent from an external point is not necessarily equal to the radius; they are independent.
2. Evaluate Reason (R): This is a standard theorem stating that the tangent at any point of a circle is perpendicular to the radius through the point of contact.
3. Conclusion: Assertion is false, but Reason is true.

Derivation:

A is false, R is true. Therefore, (D) is the correct option.

Circles — Tangents to a circle

- 38. 38.** Assertion (A): If a quadrilateral ABCD circumscribes a circle, then $AB + CD = AD + BC$. Reason (R): The lengths of tangents drawn from an external point to a circle are equal.

Answer: (A)

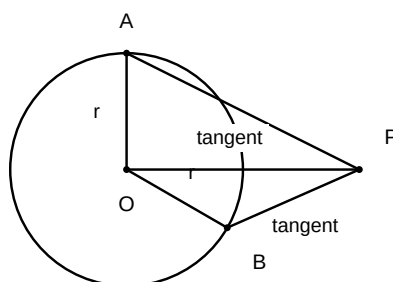
Solution:

1. Analyze Reason (R): This is the fundamental theorem that tangents from an external point to a circle are equal in length.
2. Analyze Assertion (A): Using the theorem in R, let the contact points be P, Q, R, S on sides AB, BC, CD, DA. Then $AP = AS, BP = BQ, CQ = CR, DR = DS$. Summing these shows $AB + CD = AD + BC$.
3. Conclusion: Both are true and R explains A.

Derivation:

$$\begin{aligned} AP &= AS, BP = BQ, CQ = CR, DR = DS \\ AP + BP + CR + DR &= AS + DS + BQ + CQ \\ AB + CD &= AD + BC \end{aligned}$$

- 39. 39.** Prove that the lengths of tangents drawn from an external point to a circle are equal.



Answer:

The lengths of tangents PA and PB are equal as $\triangle OAP \cong \triangle OBP$ by RHS congruence.

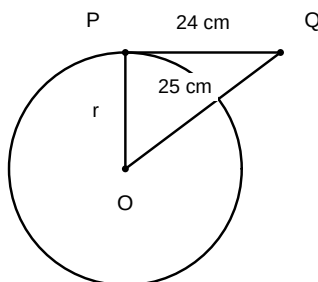
Solution:

1. Draw a circle with center O and an external point P .
2. Let PA and PB be two tangents from P touching the circle at A and B respectively.
3. Join OA , OB , and OP .
4. In $\triangle OAP$ and $\triangle OBP$, observe that $OA = OB$ (radii of the same circle).
5. $OP = OP$ (common side).
6. $\angle OAP = \angle OBP = 90^\circ$ (radius is perpendicular to tangent at point of contact).
7. By RHS congruence criterion, $\triangle OAP \cong \triangle OBP$.
8. Hence, $PA = PB$ by CPCT.

Derivation: In $\triangle OAP$ and $\triangle OBP$, $OA = OB$ (radii), $OP = OP$ (common), $\angle OAP = \angle OBP = 90^\circ$ (tangent $\perp$ radius), $\therefore \triangle OAP \cong \triangle OBP$ (RHS), $\therefore PA = PB$ (CPCT)

Circles — Tangents to a circle

- 40. 40.** From a point Q , the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. Find the radius of the circle.



Answer:

$$7 \text{ cm}$$

Solution:

1. Let O be the center of the circle and P be the point of contact.
2. We are given $OQ = 25 \text{ cm}$ and $PQ = 24 \text{ cm}$.
3. Since the radius is perpendicular to the tangent at the point of contact, $\triangle OPQ$ is a right-angled triangle with $\angle OPQ = 90^\circ$.
4. Using Pythagoras theorem: $OQ^2 = OP^2 + PQ^2$.
5. $25^2 = OP^2 + 24^2$.
6. $625 = OP^2 + 576$.
7. $OP^2 = 625 - 576 = 49$.
8. $OP = 7 \text{ cm}$.

Derivation:

$$OP^2 + PQ^2 = OQ^2$$

$$OP^2 + 24^2 = 25^2$$

$$OP^2 + 576 = 625$$

$$OP^2 = 49$$

$$OP = 7 \text{ cm}$$

Circles — Tangents to a circle